1. Creación de un nuevo proyecto en Flutter

Desde Docker, se accede a la carpeta donde se ubicarán los proyectos y se escribe:

```
flutter create app_widgets
```

2. Dejamos la cáscara en lib/main.dart

```
import 'package:flutter/material.dart';
void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  const MyApp({super.key});

  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return MaterialApp();
  }
}
```

3. Text

Veremos algunas configuraciones interesantes para Text.

```
class MyApp extends StatelessWidget {
  const MyApp({super.key});

  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: Center(
        child: Text(
          'Home',
          style: TextStyle(
```

```
            fontSize: 30,
            decoration: TextDecoration.none,
            color: Colors.red,
          ), // textStyle
        ),  // text
      ), // center
    );  // materialApp
  } // build
}
```

Para organizar nuestro código, crearemos una carpeta **widgets** dentro de **lib** y tendremos nuestras clases.

lib/widgets/text.dart

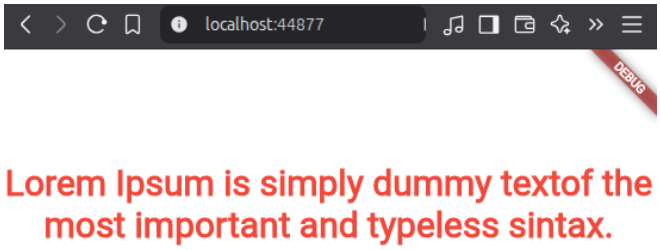
```
import 'package:flutter/material.dart';
class MyText extends StatelessWidget{
       @override
       Widget build(BuildContext context){
              return Center(
                     child: Text(
                         'Lorem Ipsum is simply dummy text '
                         'of the most important and typeless sintax.',
                         maxLines: 2,
                         overflow: TextOverflow.ellipsis,
                         style: TextStyle(
                           fontSize: 30,
                           decoration: TextDecoration.none,
                           color: Colors.red,
                         ), // textStyle
                         textAlign: TextAlign.center,
                    ),  // text
              ); // center
       }
}
```

lib/main.dart

```
import 'package:flutter/material.dart';
import 'package:app_widgets/widgets/text.dart';
void main() {
  runApp(MyApp());
}
```

```
class MyApp extends StatelessWidget {
  MyApp({super.key});

  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: MyText()
    );  // materialApp
  } // build
}
```



4. Container

También podemos tener Container que es un contenedor de widgets.

lib/widgets/container.dart

```
import 'package:flutter/material.dart';
class MyContainer extends StatelessWidget{
       @override
       Widget build(BuildContext context){
              return Center(
                     child: Container(
                            alignment: Alignment.centerLeft,
                            child: Text(
                                   'My container',
                                   style: TextStyle(
                                  fontSize: 20,
                                  decoration: TextDecoration.none,
                                  color: Colors.blue,
                                   ), // textStyle
                            ), // text
                            height: 300,
                            width: 300,
```

```
                        color: Colors.white,
                    ), // container
            ); // center
        }
}
```

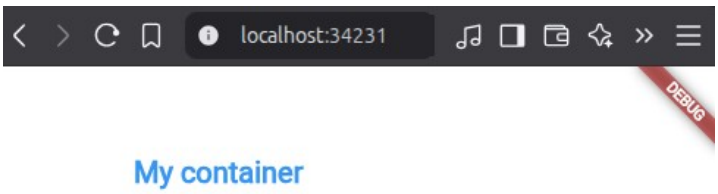
lib/main.dart

```
import 'package:flutter/material.dart';
import 'package:app_widgets/widgets/container.dart';

void main() {
  runApp(MyApp());
}

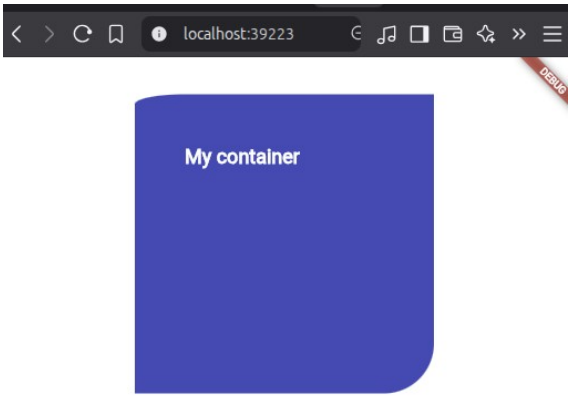
class MyApp extends StatelessWidget {
  MyApp({super.key});

  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: MyContainer()
    );  // materialApp
  } // build
}
```




```
import 'package:flutter/material.dart';
class MyContainer extends StatelessWidget{
        @override
        Widget build(BuildContext context){
                return Center(
                        child: Container(
                                alignment: Alignment.topLeft,
                                decoration: BoxDecoration(
```

```
                                color: Colors.indigo,
                                borderRadius: BorderRadius.only(
                                      topLeft: Radius.elliptical(50,10),
                                      bottomRight: Radius.circular(50),
                                ), // borderRadius
                          ),    // boxDecoration
                          child: Text(
                                'My container',
                                style: TextStyle(
                               fontSize: 20,
                               decoration: TextDecoration.none,
                               color: Colors.white,
                                ), // textStyle
                          ), // text
                          padding: EdgeInsets.all(50),
                          height: 300,
                          width: 300,
                    ), // container
          ); // center
      }
}
```



5. Column

Para organizar texto, una alternativa es usar el widget Column.

lib/widgets/column.dart

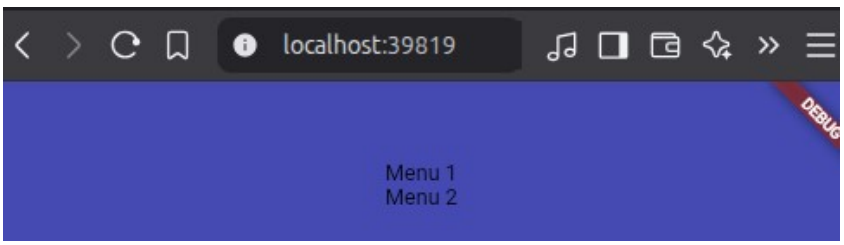
```
import 'package:flutter/material.dart';
class MyColumn extends StatelessWidget{
      @override
      Widget build(BuildContext context){
```

```
            return DefaultTextStyle(
                  style: TextStyle(
                  ), // textstyle
                  child: Container(
                        color: Colors.indigo,
                        padding: EdgeInsets.symmetric(vertical: 50),
                        child: Column(
                              children: [
                                    Text('Menu 1'),
                                    Text('Menu 2'),
                              ], // children
                        ), // column
                  ), // container
            ); // defaulttextstyle
      }
}
```

lib/main.dart

```
import 'package:flutter/material.dart';
import 'package:app_widgets/widgets/column.dart';
void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  MyApp({super.key});
  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      home: MyColumn()
    );  // materialApp
  } // build
}
```



5. Navigator (+ListView)

Comenzaremos con una simple app con dos pantallas. La primera con un ListView y la segunda con el detalle. Para movernos entre pantallas usaremos Navigator. Arranquemos con la versión preliminar.

lib/screens/noticias_scren.dart

```
import 'package:flutter/material.dart';
class NoticiasScreen extends StatelessWidget {
  NoticiasScreen({super.key});

  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return Scaffold(
       appBar: AppBar(
             title: Text('Noticias'),
       ), // appbar
       body: Center(
         child: Text(
           'Bienvenido a la seccion de noticias',
           style: TextStyle(
             fontSize: 24,
           ), // textstyle
         ), // text
       ), // center
    );  // scaffold
  } // build
}
```

lib/screens/list_view_screen.dart

```
import 'package:flutter/material.dart';
import 'package:app_widgets/screens/noticias_screen.dart';

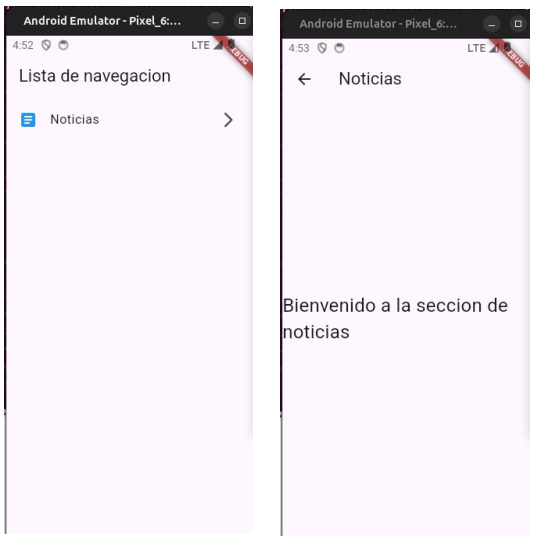
class ListViewScreen extends StatelessWidget {
  ListViewScreen({super.key});

  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return Scaffold(
       appBar: AppBar(
```

```
            title: Text('Lista de navegacion'),
      ), // appbar
      body: ListView(
            children: [
                  ListTile(
                        leading: Icon(Icons.article, color: Colors.blue),
                        title: Text('Noticias'),
                        trailing: Icon(Icons.arrow_forward_ios_rounded),
                        onTap:(){
                              Navigator.pushNamed(context, '/noticias');
                        },
                  ),
            ],
      ), // listview
    );  // scaffold
  } // build
}
```

lib/main.dart

```
import 'package:flutter/material.dart';
import 'package:app_widgets/screens/list_view_screen.dart';
import 'package:app_widgets/screens/noticias_screen.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  MyApp({super.key});
  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Lista de navegacion',
      initialRoute: '/',
      routes: {
        '/': (context) => ListViewScreen(),
        '/noticias': (context) => NoticiasScreen(),
      },
    );  // materialApp
  } // build
}
```



Ahora bien, falta incorporar el Navigator Pop que elimina la pantalla actual y regresa a la anterior. Aunque el maybePop es más robusto.

```
class NoticiasScreen extends StatelessWidget {
  NoticiasScreen({super.key});

  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return Scaffold(
       appBar: AppBar(
             title: Text('Noticias'),
       ), // appbar
       body: Padding(
         padding: EdgeInsets.all(16.0),
         child: Column(
           mainAxisAlignment: MainAxisAlignment.center,
           children: [
             Text(
               'Bienvenido a la seccion noticias',
               style: TextStyle(
                 fontSize: 24,
               ), // textstyle
             ), // text
             SizedBox(
               height: 24,
             ), // sizedbox
             ElevatedButton(
```

```
              onPressed: () {
                Navigator.pop(context);
              },
              child: Text('Volver usando pop'),
            ), // elevatebutton

            ElevatedButton(
              onPressed: () {
                if (Navigator.canPop(context)){
                  Navigator.maybePop(context);
                } else {
                  ScaffoldMessenger.of(context).showSnackBar(
                    SnackBar(
                      content: Text('No hay pantalla para volver')
                    )
                  );
                }
              },
              child: Text('Volver usando maybepop'),
            ), // elevatebutton

          ], // children
        ), // column
      ),  // padding
    );  // scaffold
  } // build
}
```



En el siguiente ejemplo pasaremos info entre pantallas. Usaremos la librería **faker** que debe ser agregada en pubspec.yaml en nuestro proyecto.

```
dependencies:
  flutter:
    sdk: flutter
faker: ^2.2.0
```

lib/screens/list_view_screen.dart

```
import 'package:flutter/material.dart';
import 'package:faker/faker.dart';
import 'package:app_widgets/screens/noticias_screen.dart';

class ListViewScreen extends StatelessWidget {
  ListViewScreen({super.key});

  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
      final faker = Faker();
    return Scaffold(
      appBar: AppBar(
            title: Text('Lista de navegacion'),
      ), // appbar
      body: ListView.builder(
            itemCount: 10,
            itemBuilder: (context, index) {
                  final title = faker.lorem.sentence();
                  final description = faker.lorem.word();

                  return ListTile(
                        leading: Icon(
                              Icons.article,
                              color: Colors.blue,
                        ), // icon
                        title: Text(title),
                        trailing: Icon(
                              Icons.arrow_forward_ios,
                              color: Colors.blue,
                        ), // icon
                        onTap: (){
                              Navigator.push(
                                    context,
                                    MaterialPageRoute(
```

```
                                                builder: (context) => NoticiasScreen(
                                                    title: title,
                                                    description: description,
                                                ),
                                            )
                                    );
                            }
                    ); // listtile
            }, // itembuilder
        ), // listview
    );  // scaffold
  } // build
}
```

lib/screens/noticias_screen.dart

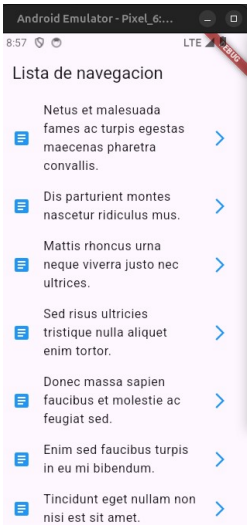
```
class NoticiasScreen extends StatelessWidget {
  final String title;
  final String description;
  NoticiasScreen({super.key, required this.title, required this.description});

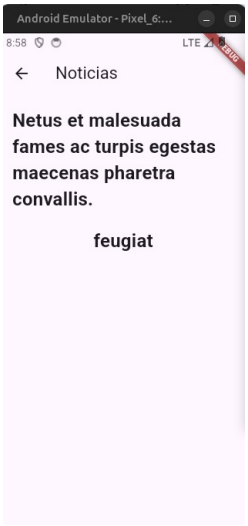
  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: Text('Noticias'),
      ), // appbar
      body: Padding(
        padding: EdgeInsets.all(16.0),
        child: Column(
          children: [
            Text(
              title,
              style: TextStyle(
                fontSize: 24,
                fontWeight: FontWeight.bold,
              ), // textstyle
            ), // text
            SizedBox(height: 20),
            Text(
              description,
```

```
              style: TextStyle(
                fontSize: 24,
                fontWeight: FontWeight.bold,
              ), // textstyle
            ), // text
          ], // children
        ), // column
      ),  // padding
    );  // scaffold
  } // build
}
```

lib/main.dart

```
class MyApp extends StatelessWidget {
  MyApp({super.key});
  // This widget is the root of your application.
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Lista de navegacion',
      initialRoute: '/',
      routes: {
        '/': (context) => ListViewScreen(),
        //'/noticias': (context) => NoticiasScreen(),
      },
    );  // materialApp
  } // build
}
```





6. Inputs

El widget TextField nos permite ingresar datos a la app. Se necesita un controlador para capturar el datos cargado. Veremos esto en el siguiente ejemplo.

lib/widgets/home_screen.dart

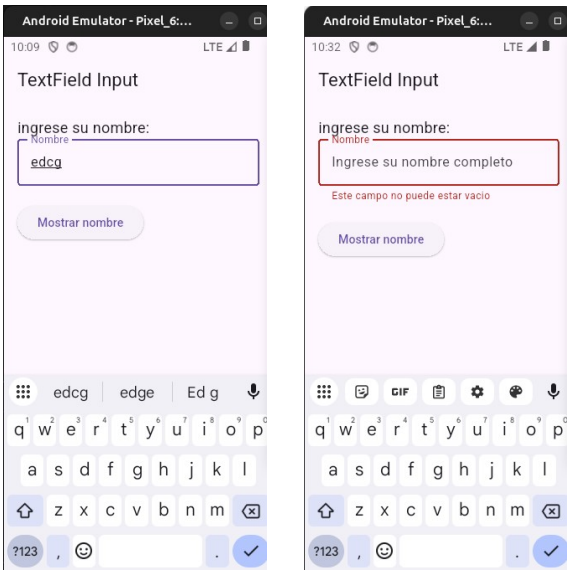
```
import 'package:flutter/material.dart';
class HomeScreen extends StatefulWidget{
  HomeScreen({super.key});
  @override
  State<HomeScreen> createState() => _HomeScreenState();
}

class _HomeScreenState extends State<HomeScreen>{
final TextEditingController _controller = TextEditingController();
  Widget build(BuildContext context) {
    return Scaffold(
       appBar: AppBar(
              title: Text(
                     'TextField Input'
              ),
       ), // appbar
       body: Padding(
              padding: EdgeInsets.all(16.0),
              child: Column(
                     crossAxisAlignment: CrossAxisAlignment.start,
                     children: [
                            Text(
                                   'ingrese su nombre: ',
                                   style: TextStyle(
                                          fontSize: 18,
                                   ),
                            ),
                            TextField(
                                   controller: _controller,
                                   decoration: InputDecoration(
                                          border: OutlineInputBorder(),
                                          labelText: 'Nombre',
                                          hintText: 'Ingrese su nombre completo',
                                   ), // inputdecoration
                                   onChanged: (text){
```

```
                                    print('Texto cambiado $text');
                              },
                        ), // textfield
                        SizedBox(height: 20),
                        ElevatedButton(
                              onPressed: () {
                                    print('nombre completo $
{_controller.text}');
                                    ScaffoldMessenger.of(context).showSnackBar(
                                          SnackBar(content:
                                                Text('Hola, $
{_controller.text}!')));
                              },
                              child: Text(
                                    'Mostrar nombre',
                              ),
                        ),    // elevatedbutton
                  ], // children
            ), // column
      ), // padding
    );  // scaffold
  } // build
}
```

lib/main.dart

```
import 'package:flutter/material.dart';
import 'package:app_widgets/widgets/home_screen.dart';
void main() {
  runApp(MyApp());
}
class MyApp extends StatelessWidget {
  MyApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'Flutter App',
      debugShowCheckedModeBanner: false,
      home: HomeScreen()
    );  // materialApp
  } // build
}
```



Aplicaremos una pequeña validación y luego usaremos checkboxes.

```
class _HomeScreenState extends State<HomeScreen>{
  final TextEditingController _controller = TextEditingController();
  String? _errorMessage;
  void _validarInput(){
      setState((){
            if (_controller.text.isEmpty){
                  _errorMessage = 'Este campo no puede estar vacio';
            } else {
                  _errorMessage = null;
            }
      });
  }

  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
            title: Text(
                  'TextField Input'
            ),
      ), // appbar
      body: Padding(
            padding: EdgeInsets.all(16.0),
            child: Column(
                  crossAxisAlignment: CrossAxisAlignment.start,
                  children: [
```

```
                        Text(
                                'ingrese su nombre: ',
                                style: TextStyle(
                                        fontSize: 18,
                                ),
                        ),
                        TextField(
                                controller: _controller,
                                decoration: InputDecoration(
                                        border: OutlineInputBorder(),
                                        labelText: 'Nombre',
                                        hintText: 'Ingrese su nombre completo',
                                        errorText: _errorMessage
                                ), // inputdecoration
                                onChanged: (text){
                                        print('Texto cambiado $text');
                                },
                        ), // textfield

                        SizedBox(height: 20),
                        ElevatedButton(
                                onPressed: () {
                                        _validarInput();
                                        if (_errorMessage == null){
                                ScaffoldMessenger.of(context).showSnackBar(
                                                        SnackBar(content:
                                                                Text('Hola, ${_controller.text}!')));
                                        }
                                },
                                child: Text(
                                        'Mostrar nombre',
                                ),
                        ),      // elevatedbutton
                ], // children
        ), // column
    ), // padding
  );  // scaffold
 } // build
}
```

7. Checkbox

Agregaremos un checkbox a la pantalla que incluía un nombre. Se usará setState() para guardar su estado y un Text para mostrar si está activo el mismo.

```
class _HomeScreenState extends State<HomeScreen>{
  final TextEditingController _controller = TextEditingController();
  bool _isAccept = false;
  String? _errorMessage;

  void _validarInput(){
       setState((){
             if (_controller.text.isEmpty){
                   _errorMessage = 'Este campo no puede estar vacio';
             } else {
                   _errorMessage = null;
             }
       });
  }
  Widget build(BuildContext context) {
    return Scaffold(
       appBar: AppBar(
             title: Text(
                   'TextField Input'
             ),
       ), // appbar
       body: Padding(
             padding: EdgeInsets.all(16.0),
             child: Column(
                   crossAxisAlignment: CrossAxisAlignment.start,
                   children: [
                         Text(
                               'ingrese su nombre: ',
                               style: TextStyle(
                                     fontSize: 18,
                               ),
                         ),
                         TextField(
                               controller: _controller,
                               decoration: InputDecoration(
                                     border: OutlineInputBorder(),
                                     labelText: 'Nombre',
```

```
                                hintText: 'Ingrese su nombre completo',
                                errorText: _errorMessage
                        ), // inputdecoration
                        onChanged: (text){
                                print('Texto cambiado $text');
                        },
                ), // textfield
                SizedBox(height: 20),
                Row(
                        children: [
                                Checkbox(
                                value: _isAccept,
                                onChanged: (bool? value){
                                        setState((){
                                                print(_isAccept);
                                                _isAccept = value ??
false;
                                        }); // setstate
                                } // onchanged
                                ), // checkbox
                                Text(
                                'Ud es profesional? '
                                ),
                        ], // children
                ),      // row
                ElevatedButton(
                        onPressed: () {
                                _validarInput();
                                if (_errorMessage == null){

        ScaffoldMessenger.of(context).showSnackBar(
                                        SnackBar(content:
                                                Text('Hola, $
{_controller.text}!')));
                                }
                        },
                        child: Text(
                                'Mostrar nombre',
                        ),
                ),      // elevatedbutton

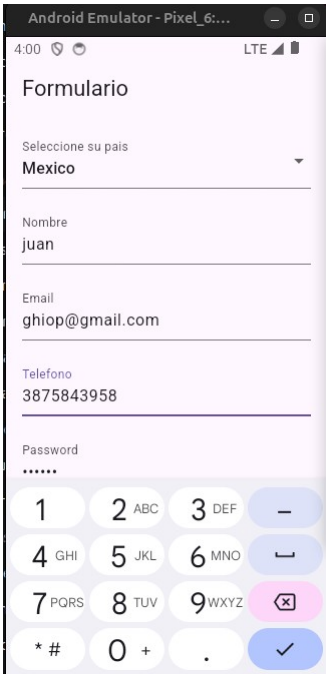
                if (_isAccept)
                        Text(
```

```
                                        ' Ud es profesional ',
                                      style: TextStyle(color: Colors.green),
                                    ), // text
                    ], // children
                ), // column
            ), // padding
        );  // scaffold
    } // build
}
```



8. Formularios

Ahora trabajaremos con formularios incluyendo texts, checkboxes y dropdowns.



lib/widgets/forms_screen.dart

```
import 'package:flutter/material.dart';
class FormsScreen extends StatefulWidget{
  FormsScreen({super.key});
  @override
  State<FormsScreen> createState() => _FormsScreenState();
}

class _FormsScreenState extends State<FormsScreen>{
  final _formKey = GlobalKey<FormState>();
  final TextEditingController _nameController = TextEditingController();
  final TextEditingController _emailController = TextEditingController();
  final TextEditingController _phoneController = TextEditingController();
  final TextEditingController _passwordController = TextEditingController();

  bool _acceptTerms = false;
  String? _selectedCountry;
  List<String> countries = ['Argentina', 'Mexico', 'Peru'];
  @override
  Widget build(BuildContext context) {
    return Scaffold(
        appBar: AppBar(
                title: Text(
                        'Formulario'
                ),
        ), // appbar
        body: SingleChildScrollView(
                padding: EdgeInsets.all(16.0),
                child: Form(
                        key: _formKey,
                        child: Column(
                                crossAxisAlignment: CrossAxisAlignment.start,
                                children: <Widget>[
                                        DropdownButtonFormField<String>(
                                                decoration: InputDecoration(
                                                        labelText: 'Seleccione su pais'
                                                ),
                                                value: _selectedCountry,
                                        items: countries.map((String country){
                                                return DropdownMenuItem(
                                                        value: country,
                                                        child: Text(country)
                                                ); // DropdownMenuItem
                                        }).toList(),
```

```
                                        onChanged: (String? newValue){
                                                setState((){
                                                        _selectedCountry = newValue;
                                                });
                                        },
                                        validator: (value){
                                                if (value == null || value.isEmpty){
                                                        return 'por favor ingrese un pais';
                                                }
                                                return null;
                                        },
                                        ), // dropdown
                                        SizedBox(height: 20),
                                        TextFormField(
                                                controller: _nameController,
                                                decoration: InputDecoration(
                                                        labelText: 'Nombre',
                                                        hintText: ' Ingrese su nombre: ',
                                                ),
                                                validator : (value) {
                                                        if (value == null || value.isEmpty){
                                                                return 'por favor ingrese su nombre
';
                                                        }
                                                        return null;
                                                }
                                        ), // textformfield
                                        SizedBox(height: 20),
                                        TextFormField(
                                                controller: _emailController,
                                                keyboardType: TextInputType.emailAddress,
                                                decoration: InputDecoration(
                                                        labelText: 'Email',
                                                        hintText: ' Ingrese su email: ',
                                                ),
                                                validator: (value) {
                                                        if (value == null || value.isEmpty){
                                                                return 'por favor ingrese su email
';
                                                        }
                                                        if (!RegExp(r'^[^@]+@[^@]+\.[^@]
+').hasMatch(value)){
                                                                return 'ingrese correo valido';
                                                        }
                                                        return null;
                                                },
```

```
                                ), // textformfield
                                SizedBox(height: 20),
                                TextFormField(
                                        controller: _phoneController,
                                        keyboardType: TextInputType.phone,
                                        decoration: InputDecoration(
                                                labelText: 'Telefono',
                                                hintText: ' Ingrese su telefono: ',
                                        ),
                                        validator: (value) {
                                                if (value == null || value.isEmpty){
                                                        return 'por favor ingrese su
telefono ';
                                                }
                                                if (!RegExp(r'^[0-9]+$').hasMatch(value)){
                                                        return 'ingrese telefono valido';
                                                }
                                                if (value.length!=10){
                                                        return 'El numero debe tener 10
digitos';
                                                }
                                                return null;
                                        },
                                ), // textformfield
                                SizedBox(height: 20),
                                TextFormField(
                                        controller: _passwordController,
                                        obscureText: true, // para que no se vea
                                        keyboardType: TextInputType.phone,
                                        decoration: InputDecoration(
                                                labelText: 'Password',
                                                hintText: ' Ingrese su password: ',
                                        ),
                                        validator: (value) {
                                                if (value == null || value.isEmpty){
                                                        return 'por favor ingrese su
password ';
                                                }
                                                if (value.length<6){
                                                        return 'ingrese contraseña valida';
                                                }
                                                return null;
                                        },
                                ), // textformfield
                                SizedBox(height: 20),
                                Row(
```

```
                                        children: [
                                                Checkbox(
                                                        value: _acceptTerms,
                                                        onChanged: (bool? value){
                                                                setState((){
                                                                        _acceptTerms = value
?? false;
                                                                });
                                                        },
                                                ), // checkbox
                                                Text('Acepto los terminos y condiciones'),
                                        ], // children
                                ),  // row
                                if(!_acceptTerms)
                                        Text(
                                                'Debes aceptar los terminos y
condiciones',
                                                style: TextStyle(color: Colors.red),
                                        ), // text
                                SizedBox(height: 20),
                                ElevatedButton(
                                        onPressed: (){
                                                if (_formKey.currentState!.validate() &&
_acceptTerms){

        ScaffoldMessenger.of(context).showSnackBar(SnackBar(
                                                        content: Text('Formulario
valido'),
                                                        )); // snackbar
                                                } else if (!_acceptTerms){
                                                        setState(() {});
                                                }
                                        },
                                        child: Text('Enviar'),
                                ), // elevatedButton
                        ], // children
                ), // column
        ), // form
        ), // singleScrollView
    );  // scaffold
  } // build
}
```

8.5 Refactorización

Para que sea mantenible y reusable en el tiempo, es necesario que el código esté refactorizado.

En la carpeta en DRIVE se encuentra la refactorización del atributo de **nombre**.

Este se debe hacer con el resto de los widgets del form.

Una alternativa para seguir refactorizando es usar **Providers**. De esta manera todos las validaciones van en una clase (además de que los atributos del form, cada uno van en clase separada). De esta manera, tenemos un código mantenible, reusable y más fácil de entender.